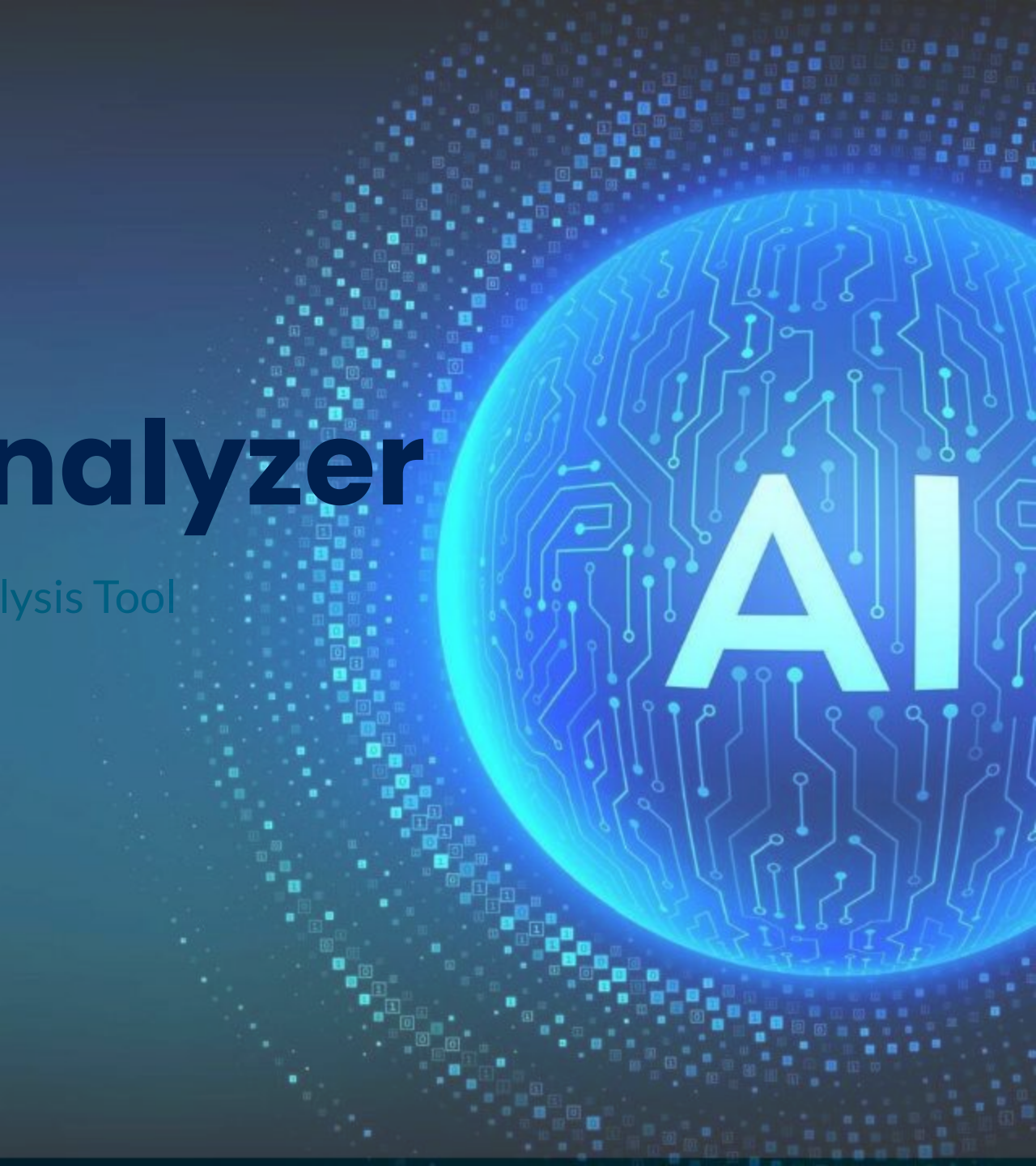


Exam Paper Analyzer

MOE O-Level History Specimen Paper Analysis Tool

Candidate: Ng Wei Ern



PROBLEM STATEMENT

100%
MANUAL REVIEW

Efficiency & Verification Gaps

MOE educators currently verify exam papers manually—a process that is often time-consuming, subjective, and lacks a rigorous audit trail.

- ✕ Are all syllabus objectives covered?
- ✕ Is the mark allocation balanced across topics?
- ✕ Is the cognitive demand level-appropriate?

THE PROPOSED SOLUTION

AI-Powered Semantic Insight

An intelligent tool designed to analyze any exam PDF against a syllabus PDF, producing structured alignment reports in seconds.



LLM-Based Analysis: Understands intent, not just keywords.



Automated Metrics: Alignment and balance scoring.



Bloom's Taxonomy: Automatic cognitive tagging.

ARCHITECTURE AND DESIGN



Web Interface (main.py)

The entry point and routing layer. It orchestrates backend logic and handles incoming user requests seamlessly.



Core Logic (analyzer.py)

The heavy lifting: this module processes and analyzes the semantic content of exams using complex AI models.



Evaluation (evaluate.py)

Manages scoring logic and metrics, ensuring that the processed exams meet predefined pedagogical standards.



Presentation (templates/)

Dynamic UI driven by Jinja2 templates (index/results.html) for responsive result visualization.

TECHNICAL STACK



FastAPI & Uvicorn

High-performance asynchronous handling for heavy document parsing and API operations.



OpenAI & Anthropic

Deep semantic analysis leveraging Claude Sonnet and GPT-4o Mini for intelligent interpretation.



PDF Processing

Robust libraries: pdfplumber, PyPDF2, and Tesseract for extracting text and structured data.



Frontend Rendering

Jinja2 server-side templating paired with Chart.js for interactive data visualization.

SYSTEM DEMO OVERVIEW

From Document to Dashboard

The tool ingests high-level PDF inputs and converts them into actionable data visualizations.



Input: Syllabus 2174 & Specimen Paper.



Processing: PDF Extraction -> LLM Analysis.



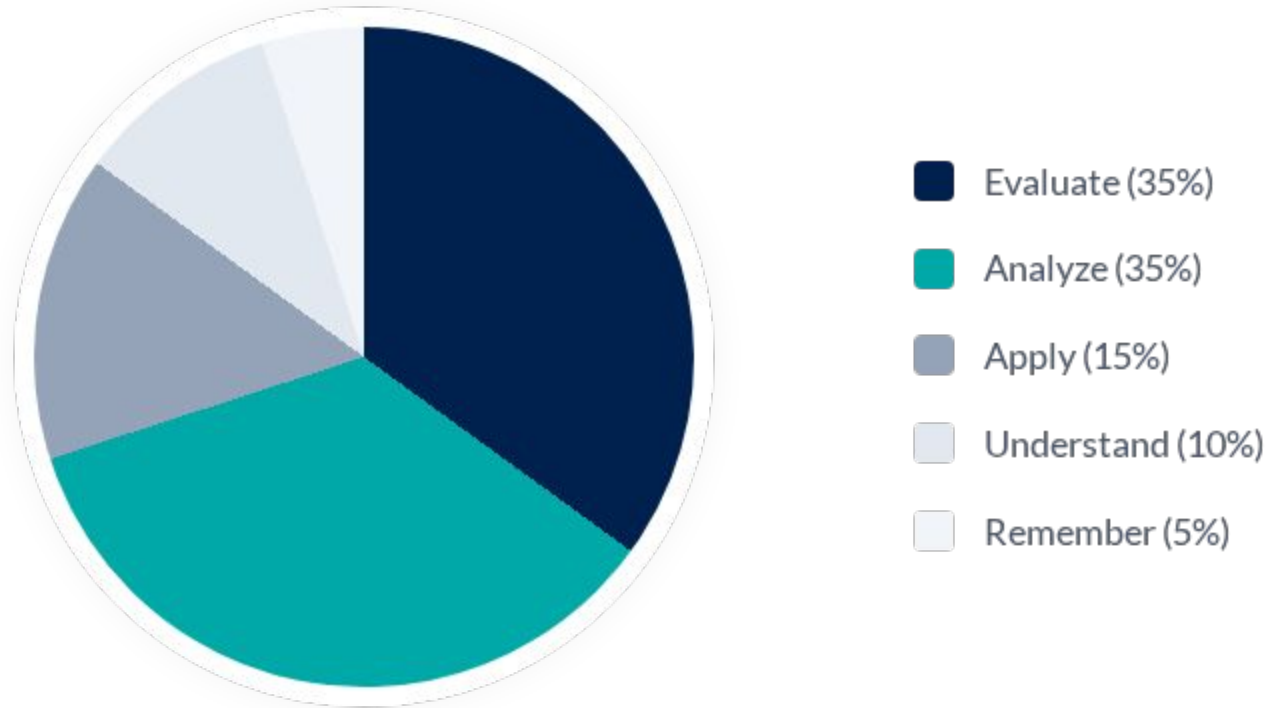
Output: Scoring, Weightage, & Recommendations.

KEY PERFORMANCE SCORES



Analysis based on O-Level History 2174/01. Scores reflect high AO2/AO3 coverage with minor gaps in local resistance narratives.

COGNITIVE DEMAND BREAKDOWN



High-order thinking emphasis appropriately weighted for GCE O-Level.

EVALUATION FRAMEWORK

Metric	Result	Interpretation
Marks Accuracy	100%	Predicted marks match ground truth exactly
Bloom's Exact Agreement	78%	7/9 questions correctly tagged
Covered Objectives Recall	80%	Most syllabus objectives identified
Factual Consistency		Internal logic and totals are consistent

Overall Performance Score: 89.5/100

EVALUATION DESIGN CHOICES

-  **Soft/Fuzzy Matching for Objectives:** Exact string matching misses paraphrasing. We use pragmatic overlap (≥ 2 common words) to identify correctly identified objectives.
-  **Adjacent Bloom's Agreement:** Since "Analyze" and "Evaluate" are often ambiguous for humans, adjacent level agreement is measured to reflect near-misses.
-  **Addressing Hallucinations:** Consistency checks ensure totals and percentages are factually grounded, preventing wild AI interpretations.

ROADMAP & NEXT STEPS



Short Term

Build labelled dataset of 20 past papers. A/B test model choices (Sonnet vs GPT-4o).



Medium Term

Batch analysis to track alignment trends over years. Add PDF export features.



Integration

Connect with MOE's existing curriculum management systems via JSON API.



Production

Deploy on GovTech-approved cloud with full data residency and audit trails.

Q&A

Augmentation, not Replacement.

The tool handles tedious extraction; educators apply domain judgment. Surfacing information for human review is the primary goal.